

$$\theta = \max(P(x + 1 | x) - P(x | x - 1))$$

1. $P(x + 1 | x)$: - This is the conditional probability of adding one more pellet given that there are already x pellets in the cluster. It represents the likelihood that the mice will add another pellet to the cluster of size x .

2. $P(x | x - 1)$: - This is the conditional probability of having x pellets given that there were $x - 1$ pellets in the cluster. It represents the likelihood that the mice will add a pellet to the cluster of size $x - 1$.

3. $P(x + 1 | x) - P(x | x - 1)$: - This difference measures the change in the likelihood of adding another pellet between two successive cluster sizes. A larger difference indicates a more significant change in behavior.

4. $\max(P(x + 1 | x) - P(x | x - 1))$: - The maximum value of the differences calculated above. This identifies the point where the change in behavior is most pronounced, which can be interpreted as the threshold where the mice are most likely to stop adding more pellets to their cluster.

In summary, the formula aims to find the point where the mice's behavior changes the most significantly in terms of adding pellets to their cluster. This can help identify when the mice "give up" on adding more pellets.